Deciphering hypothalamic pathology in ALS: insights into non-motor symptoms and disease progression

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Summary

Amyotrophic lateral sclerosis (ALS) is a fatal neurodegenerative disease with highly heterogeneous clinical presentations. Among the non-motor features increasingly recognised as clinically relevant is the dysregulation of energy balance, with weight loss—particularly due to fat mass loss—emerging as a significant modifier of disease risk, progression, and survival. In this context, the hypothalamus—a key regulator of homeostatic and metabolic processes—has gained attention for its potential role in ALS pathophysiology.

This Review synthesises emerging evidence of hypothalamic involvement in ALS, including neuronal loss, proteinopathy, and volume loss observed through histological and neuroimaging studies. We critically examine current imaging approaches and their technical limitations and explore neuroendocrine dysfunction across the hypothalamic-pituitary axes. Collectively, these findings suggest that hypothalamic dysfunction may contribute to clinically relevant metabolic, sleep, behavioural, and cognitive changes in ALS, adding to our understanding of ALS as a multisystem disease. Continued investigation of the hypothalamus may reveal novel biomarkers, inform risk stratification, and identify therapeutic opportunities to address disease heterogeneity and improve clinical outcomes.

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Introduction

Motor neuron disease (MND) is a progressive neurodegenerative disorder characterised by the degeneration and death of upper motor neurons in the motor cortex and brainstem nuclei, and/or lower motor neurons in the spinal cord.[1] MND encompasses a spectrum of clinically distinct subtypes that differ in their site of onset, severity, and rate of progression. Primary lateral sclerosis (PLS) is characterised by isolated upper motor neuron signs, with diagnosis typically confirmed only after a prolonged absence of lower motor neuron involvement. In contrast, progressive muscular atrophy (PMA) presents with predominant lower motor neuron features, although some patients may later develop upper motor neuron signs.[1] These subtypes are associated with distinct clinical trajectories and prognoses. Amyotrophic lateral sclerosis (ALS), the most common and widely recognised form of MND, is defined by the combined presence of upper and lower motor neuron signs. These typically manifest with varying severity as limb muscle weakness, dysphagia, and dysarthria.[1] Survival in ALS is variable but typically ranges from 2 to 5 years following symptom onset. While most cases are sporadic, approximately 10% are familial, with numerous genetic causes contributing to both forms of the disease.[1,2] More than 50% of patients exhibit cognitive and behavioural symptoms, and a small subset of patients meet the diagnostic criteria for ALS-frontotemporal dementia (ALS-FTD).[1,3]

Although ALS has historically been regarded as a disorder confined to the neuromotor system, accumulating evidence suggests that ALS is a multisystem disease, highlighting the relevance of non-motor features, which can significantly influence disease progression, quality of life, and survival outcomes.[1] Among these, metabolic and appetite dysregulation have emerged as consistent and clinically meaningful manifestations.[4] Weight loss is now recognised as a modifiable risk factor for ALS progression. Lower body weight often precedes symptom onset, with a lower premorbid body mass index (BMI) associated with increased ALS risk. Longitudinal studies show that weight and BMI decline can begin up to a decade before clinical onset, and are associated with increased risk of

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Key points

- Dysfunction of the hypothalamus is hypothesised to contribute to non-motor aspects of amyotrophic lateral sclerosis (ALS), leading to poorer outcomes for some people living with the disease.
- Proteinopathy and neuronal loss are commonly observed in the hypothalamus of people living with ALS, particularly in regions critical for regulating energy homeostasis and sleep.
- Magnetic resonance imaging provides *in vivo* evidence of hypothalamic volume loss and atrophy in ALS, which may disproportionately affect specific hypothalamic subregions.
- Specific neuropeptidergic neuronal populations are affected in the hypothalamus in ALS, particularly within the lateral hypothalamic area, and their involvement may contribute to metabolic impairment.
- Research to date suggests dysregulation across multiple hypothalamic–pituitary axes in ALS; however, little is known about upstream hypothalamic pathways and their roles in metabolic dysfunction and disease progression.

developing ALS and poorer clinical outcomes.[4] Conversely, adult weight gain appears to be protective. At diagnosis, lower BMI correlates with reduced survival, and both underweight and overweight individuals may experience accelerated functional decline and increased mortality.[4] Continued weight and BMI loss after diagnosis further compounds risk for faster disease progression.[4]

In ALS, weight loss arises from reductions in both lean muscle mass and fat mass. Muscle atrophy is attributable to progressive motor neuron degeneration and denervation.[5] However, the degree of fat mass loss varies considerably between individuals and appears to be a stronger predictor of clinical outcome. Greater fat mass loss has been linked with faster functional decline, accelerated disease progression, and shorter survival, while increased fat mass is associated with more favourable prognosis and reduced mortality risk.[4] These findings position fat mass not only as a marker of disease severity, but also as a potential target for therapeutic intervention.

Malnutrition in ALS, characterised in part by fat mass loss, is attributed to both reduced energy intake and hypermetabolism, the latter defined as an increase in resting energy expenditure.[4] While the risk of malnutrition in ALS has long been recognised, hypermetabolism was first identified in the 1990s in a subset of patients with elevated energy needs, and later confirmed in transgenic ALS mouse models.[4] Hypermetabolism is associated with faster disease progression and worse prognosis, although it is unclear whether it directly contributes to weight loss.[6] Inadequate dietary intake relative to elevated energy demands likely plays a greater role in fat mass depletion. In this context, appetite loss may be of greater clinical concern when considering the link between energy imbalance and disease progression. Loss of appetite is frequently observed in ALS and contributes meaningfully to ongoing weight loss. It is not universally associated with dysphagia or respiratory impairment, and is likely driven by a combination of functional, psychosocial, and physiological factors.[7] Appetite loss correlates with lower body weight and progressive fat mass loss following diagnosis, further implicating it in poorer clinical outcomes.[4]

Whole-body energy balance is centrally regulated by neural systems that dynamically manage energy intake and expenditure. Among these, the hypothalamus is a key integrator of peripheral and central signals that influence metabolic behaviour. In this Review, we examine current evidence for hypothalamic dysfunction as a contributor to metabolic and appetite dysregulation in ALS. Through a critical evaluation of histological, imaging, and endocrine studies, we explore the relevance of hypothalamic changes to clinical outcomes and assess the potential of the hypothalamus as a therapeutic target in ALS.

The role of the hypothalamus in energy homeostasis

The hypothalamus is a central integrator of homeostatic signals that regulate whole-body energy balance. It comprises interconnected nuclei that receive peripheral hormonal inputs and project to the forebrain, brainstem, and autonomic centres to coordinate feeding, metabolism, and neuroendocrine function (Fig. 1; Panel 1). Key hypothalamic regions involved in energy regulation include the arcuate nucleus (ARC), paraventricular nucleus (PVN), ventromedial hypothalamic nucleus (VMH), and lateral hypothalamic area (LHA). These nuclei are responsive to circulating signals such as leptin, ghrelin, insulin, and glucocorticoids, and are central to regulating energy intake and expenditure.[8]

The hypothalamus integrates humoural and neural inputs to regulate resting metabolic rate, thermogenesis, and appetite-related behaviours. Basal metabolic rate is modulated indirectly via the hypothalamic-pituitary-gonadal (HPG) and growth hormone-insulin-like growth factor 1 (GH-IGF-1) axes; directly by the hypothalamic-pituitary-thyroid (HPT) axis, which governs mitochondrial metabolism, and the hypothalamic-pituitary-adrenal (HPA) axis, which modulates energy substrate utilisation.[8] The preoptic area supports cold-induced thermogenesis, while ARC and VMH circuits

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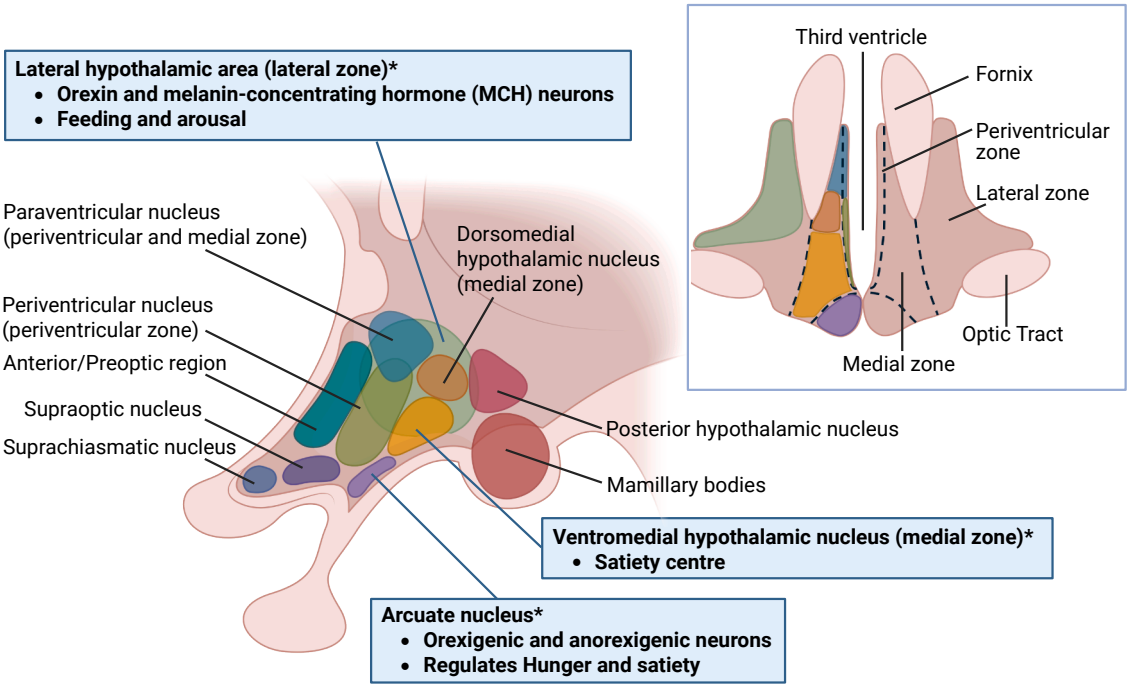

Fig. 1: Summary of hypothalamic nuclei. The main image shows all nuclei on the sagittal plane. Inset displays a coronal view at the level of the arcuate nucleus. Nuclei that are known to be involved in feeding behaviour and are hypothesised to be affected in amyotrophic lateral sclerosis (ALS) are highlighted in blue boxes.

regulate food intake through homeostatic and behavioural pathways. In parallel, hypothalamic regulation of sympathetic tone contributes to diet-induced thermogenesis, and the suprachiasmatic nucleus (SCN) coordinates circadian fluctuations in metabolism.[8]

The ARC, positioned near the median eminence, contains two key neuronal populations: orexigenic NPY/AgRP neurons that stimulate hunger and reduce energy expenditure, and anorexigenic POMC neurons that promote satiety through melanocortin receptor activation in the PVN. Peripheral signals tightly modulate this balance—leptin suppresses NPY/AgRP and activates POMC neurons, while ghrelin exerts the opposite effect. These circuits play a critical role in maintaining energy stores and responding to dynamic metabolic demands.[8] Beyond appetite regulation, the hypothalamus influences peripheral metabolism via descending autonomic pathways. The ARC-PVN circuitry modulates adaptive thermogenesis in response to energy surplus or deficit and hypothalamic outflow to brown adipose tissue contributes to non-shivering thermogenesis.[9] Disruption of these circuits in ALS may exacerbate energy imbalance and contribute to accelerated muscle wasting.

The hypothalamus is also a central target for multiple endocrine signals involved in metabolic regulation. While leptin and ghrelin have been more extensively studied in ALS, other hormones such as insulin, thyroid hormones, and glucocorticoids also influence hypothalamic activity. For example, insulin modulates hypothalamic glucose uptake and neuropeptide expression, while thyroid hormones regulate mitochondrial function and basal metabolism.[8] Glucocorticoids affect substrate partitioning and appetite through their actions on lipid and glucose metabolism. In ALS, altered levels of these hormones have been reported, though whether such changes reflect primary hypothalamic dysfunction or secondary downregulation of hormonal sensitivity—such as has been proposed for ghrelin[10]—remains unclear.

Altogether, emerging evidence suggests that hypothalamic dysfunction in ALS may contribute to disordered appetite, dysregulated energy expenditure, and body weight loss. These changes likely worsen energy imbalance and may influence disease progression and severity.

Histological evidence of hypothalamic involvement

Several studies have used immunohistochemical staining to identify neurons and pathological protein inclusions in postmortem hypothalamic tissue from people living with ALS (plwALS). With few exceptions,[11]

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Panel 1: **Anatomy of the hypothalamus.**

The hypothalamus is a small but structurally and functionally complex diencephalic structure. It lies anterior and inferior to the thalamus, forming the floor and part of the lateral walls of the third ventricle (Fig. 1). Posteriorly, it is bordered by the subthalamus; anteriorly, by the optic chiasm. The pituitary stalk projects inferiorly from the hypothalamus, connecting it to the pituitary gland. Anatomical landmarks demarcating its boundaries include the anterior commissure (rostrally), mammillary bodies (caudally), optic chiasm (ventrally), and the fornix, which traverses the central region and links the hypothalamus with the hippocampus. The third ventricle bisects the hypothalamus along the midline, dividing it into symmetrical left and right halves. The hypothalamus occupies a central position at the intersection of neuroendocrine, autonomic, limbic, and visual systems. It is this integration of these diverse pathways that underpins its essential role in regulating homeostasis and behaviour. The anatomical and functional complexity of the hypothalamus underscores its central role in integrating internal and external cues to maintain homeostasis across endocrine, autonomic, and behavioural domains.

Anatomically, the hypothalamus is subdivided rostrocaudally into three major regions:

- The *anterior (preoptic) region*, which includes the medial and lateral preoptic nuclei and the SCN, is generally involved in thermoregulation, reproductive behaviours, and circadian rhythm regulation.
- The *tuberal region*, positioned posterior to the preoptic area, contains several nuclei critical for energy balance and feeding, including the ARC, VMH, and DMH.
- The *posterior region*, which includes the mammillary bodies and posterior hypothalamic nucleus, plays roles in autonomic function and memory processing.

Transversely, the hypothalamus is organised into three zones:

- The *medial zone* houses key neuroendocrine and metabolic regulatory nuclei, including the PVN, VMH, and DMH.
- The *lateral zone* contains the LHA, a region important for arousal, feeding, and reward-related behaviours.
- The *periventricular zone*, adjacent to the third ventricle, includes nuclei such as the periventricular nucleus (PeVN) and parts of the PVN, which contribute to autonomic and neuroendocrine regulation.

This anatomical organisation reflects the hypothalamus' role as a central homeostatic regulator. The ARC, located at the base of the hypothalamus near the median eminence, is ideally positioned to sense circulating metabolic signals. It contains two functionally distinct neuronal populations: NPY/AgRP-expressing neurons, which promote hunger and reduce energy expenditure, and POMC/α-MSH-expressing neurons, which activate melanocortin receptors in the PVN to induce satiety. The PVN is a key integrator of stress responses, appetite control, and hypothalamic-pituitary axis regulation, particularly via the HPA axis.

protein inclusions have been observed throughout the hypothalamus.[12–16] Neuronal loss has also been reported across multiple hypothalamic nuclei,[13,14,17] suggesting that hypothalamic pathology may contribute to non-motor symptoms observed in ALS. A summary of histological evidence is presented in Fig. 2. In addition to human studies, several lines of evidence from ALS mouse models support hypothalamic involvement. Many of these models exhibit weight loss, reduced fat mass, and altered feeding behaviours,[18] or fail to gain body weight following disease onset.[19] Accordingly, we have considered histological evidence from both human and mouse tissues to provide a more comprehensive overview of hypothalamic pathology in ALS.

TDP-43, encoded by the *TARDBP* gene, is a nuclear RNA- and DNA-binding protein involved in transcription, splicing, and RNA stability.[20] In ALS, TDP-43 becomes mislocalised to the cytoplasm, where it aggregates into phosphorylated inclusions—a hallmark observed in up to 98% of ALS cases. Although the mechanisms remain incompletely understood, nuclear depletion and cytoplasmic aggregation of TDP-43 are thought to contribute to neurodegeneration through loss of normal function and gain of toxic function.[20] TDP-43 pathology has been detected in cortical regions associated with executive function, language, and behaviour,[21] as well as the spinal cord, brainstem,[21] cerebellum,[22] hippocampus, and striatum.[23] Within the hypothalamus, inclusions have been reported in the LHA,[22] PVN, PeVN, preoptic area,[13,15] and suprachiasmatic nucleus (SCN).[12] This is notable, as the LHA and PVN contain secondary neurons essential for regulating appetite and energy balance.[24]

TDP-43 pathology has also been observed in specific neuropeptidergic neuronal populations. In plwALS, orexin-, oxytocin-, and melanin-concentrating hormone (MCH)-expressing neurons have been shown to contain inclusions, with corresponding reductions in their number in postmortem hypothalamic tissue.[13,14] However, TDP-43 aggregates are present in only a subset of these cells, and no consistent co-localisation with specific neuronal subtypes has been identified in either human or animal studies.[13,14]

To better contextualise these histological changes, it is important to consider the anatomical localisation and functional roles of affected hypothalamic neurons. POMC- and AgRP-expressing neurons are localised to the ARC and act as important gatekeepers of the melanocortin system. POMC is a precursor of anorexigenic melanocortin peptides, while AgRP-expressing neurons

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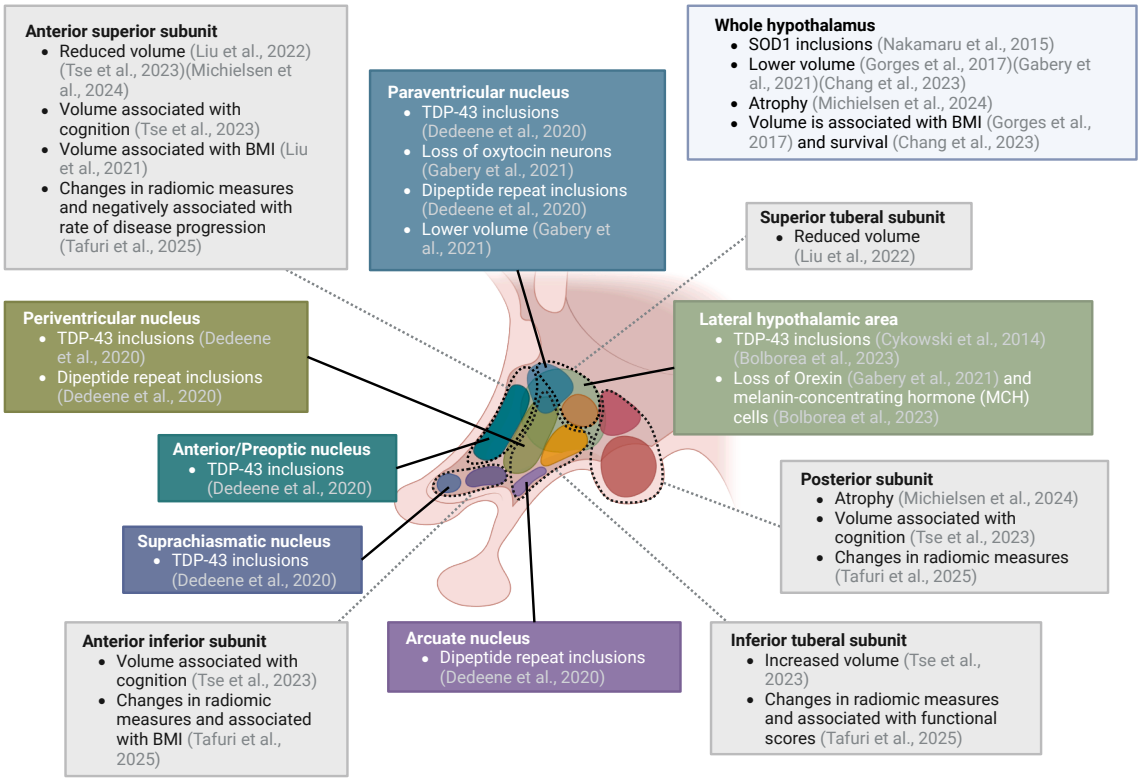

Fig. 2: Summary of evidence for hypothalamic deficits and dysfunction in people living with amyotrophic lateral sclerosis (ALS). Grey boxes describe imaging findings for hypothalamic subunits, as indicated by dashed outlines.

secrete melanocortin antagonists, promoting hunger. Localised to the LHA, MCH- and orexin-expressing neurons receive inputs from other hypothalamic nuclei, including the ARC, and are implicated in energy balance. MCH-expressing neurons stimulate appetite and suppress energy expenditure through the melanocortin system, while orexin neurons promote feeding behaviour and inhibit satiety. MCH and orexin also contribute to sleep regulation, with MCH facilitating rapid eye movement (REM) sleep and orexin maintaining wakefulness. Oxytocin neurons within the PVN play an additional role by enhancing leptin signalling to the nucleus tractus solitarius, further promoting satiety.

Importantly, many of these neuropeptidergic populations show structural alterations in ALS mouse models. In SOD1^{G86R} mice, key genes of the melanocortin pathway are dysregulated.[25] Histologically, POMC- and MCH-positive neurons are decreased, while the density of AgRP-positive fibres is increased.[13,25] These changes parallel those observed in plwALS and suggest that the melanocortin system is selectively vulnerable in ALS. Notably, intracerebroventricular administration of MCH in SOD1^{G86R} mice increases food intake, maintains body weight, and improves lower motor neuron survival.[26] Sleep alterations observed in three ALS mouse models were also rescued with either MCH or dual orexin receptor antagonist.[26] This is a compelling observation, as emerging evidence suggests orexin may influence motor control, although this has not been demonstrated in clinical populations.[27]

Further evidence comes from a mouse model of TDP-43 overexpression in wild-type FVB/N mice, in which viral vector-mediated expression of TDP-43 leads to hypothalamic atrophy and the loss of orexin-, MCH-, and oxytocin-expressing neurons.[28] These histopathological changes are accompanied by metabolic dysregulation and apathy-like behaviour, again supporting a role for hypothalamic dysfunction in ALS-related non-motor symptoms. Collectively, these animal studies demonstrate structural and functional involvement of multiple hypothalamic nuclei and neuropeptidergic systems in ALS.

In addition to TDP-43, other proteinopathies have been observed in the hypothalamus. SOD1 aggregation has been reported in one family with a truncating SOD1 mutation, although specific hypothalamic localisation was not detailed.[16] Dipeptide repeat protein pathology associated with *C9ORF72*-linked ALS has also been observed throughout the hypothalamus, including in the PVN, PeVN, preoptic area, and ARC.[12]

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Together, findings confirm that the hypothalamus is susceptible to neurodegenerative processes in ALS, including proteinopathy and neuronal loss. Notably, evidence from animal studies suggest selective vulnerability of the hypothalamic melanocortin system. Although this selective vulnerability may be observed in some plwALS, the broader interactions between affected nuclei, hormonal regulation, and metabolic consequences remain poorly understood. Future studies should aim to integrate findings across neuronal subtypes, pathogenic proteins, and hypothalamic circuits to clarify how hypothalamic dysfunction contributes to disease progression and non-motor symptomatology in ALS.

Imaging evidence of hypothalamic dysfunction

Magnetic resonance imaging (MRI) enables the characterisation of brain structures *in vivo* and has been widely used in amyotrophic lateral sclerosis (ALS) to investigate neuroanatomical changes throughout the central nervous system.[29] The hypothalamus, though challenging to image due to its small size and deep location, can now be examined using automated segmentation tools such as Freesurfer.[30] However, limitations remain, including resolution constraints and variable parcellation schemes used to define hypothalamic subregions. Panel 2 outlines some methodological considerations associated with MRI-based hypothalamic analysis. In this section, we present key imaging findings, using the same terminology employed in the literature. A visual summary is provided in Fig. 2.

With few exceptions,[40] studies consistently report lower whole and subunit hypothalamic volumes in plwALS compared with healthy controls.[38,41–45] Volume loss is also observed in presymptomatic carriers of ALS-associated genetic variants.[42] Regionally, volume reductions are most often reported in the anterior-superior[43,45,46] and superior-tuberal subunits,[43] although one study noted increased volume in the left inferior-tuberal subunit.[43] Posterior hypothalamic volume has been linked to age at disease onset,[42] and longitudinal studies suggest accelerated atrophy in this region among plwALS compared to controls.[45] Available evidence does not support consistent volumetric differences between familial and sporadic ALS.[42,47] Additional work using radiomic analyses has shown alterations in anterior, inferior-tuberal, and posterior subunits.[48]

Few studies have assessed the relationship between hypothalamic volume and appetite in ALS, although one study reported no significant association.[38] In contrast, links between hypothalamic volume and cognition have been more consistently reported. Lower volumes in anterior-superior, anterior-inferior, and posterior subunits have been associated with memory and visuospatial deficits,[45] and similar findings were observed in a broader ALS–FTD cohort.[46] Understanding how hypothalamic atrophy contributes to appetite loss, and how these changes may interact with behavioural symptoms, remains an area for further investigation. Such insights may eventually inform targeted interventions to support nutritional status in ALS.

Findings on the relationship between hypothalamic volume and BMI are mixed. One radiomics-based study identified a strong correlation between BMI and the left anterior-inferior subunit,[48] while others report broader associations between hypothalamic volume and BMI, particularly in patients with familial ALS.[38] In a cohort spanning a wide BMI range, a U-shaped association was observed between hypothalamic volume and BMI, suggesting that both low and high extremes of adiposity may reflect underlying hypothalamic dysfunction.[38] In the same study, low hypothalamic volume in combination with low BMI was strongly predictive of poorer prognosis. Larger cohort data further support an association between reduced hypothalamic volume and shorter survival,[45] while radiomic measures in the right inferior-tubular and anterior-superior subunits have been linked to ALSFRS-R scores (a commonly used functional rating scale in ALS) and the rate of disease progression.[48]

Beyond volumetrics, few imaging studies have explored functional or structural connectivity of the hypothalamus in ALS. Diffusion-weighted imaging has revealed microstructural abnormalities in the orbitofrontal-hypothalamic tract,[49] and resting-state functional MRI has demonstrated hypothalamic hyperconnectivity with the cerebellum, occipital lobe, and cingulate gyrus in familial ALS.[50] These findings suggest that hypothalamic involvement in ALS may extend beyond local atrophy, affecting large-scale neural networks with potential relevance to appetite regulation, behaviour, and homeostatic balance.

The role of the hypothalamic–pituitary axis in ALS

As noted above, the hypothalamus plays a pivotal role in coordinating multiple hormonal axes that regulate whole-body energy homeostasis, acting as a central integrator of neuroendocrine signals. Evidence of pathology along the hypothalamic-pituitary axis in ALS underscores the potential contribution of hypothalamic dysfunction to disease-related metabolic dysregulation. In addition to TDP-43 pathology within the hypothalamus itself, Dedeene and colleagues[12] identified dipeptide repeat protein inclusions in the pituitary gland of individuals with *C9ORF72*-associated ALS, including in regions associated with several hypothalamic–pituitary axes involved in metabolic control.

While histopathological and neuroimaging studies have provided important insights into hypothalamic

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Panel 2: **Limitations of current MRI-based studies.**

Imaging studies of the hypothalamus typically segment the structure using established methodologies, including manual tracing by expert raters or automated processing through specialised software. Whole-hypothalamus segmentation relies on anatomical landmarks and tissue-specific magnetic properties (Fig. 3), as demonstrated by tools such as OSHy-X (Fig. 3a).[31] Other approaches focus on subdividing the hypothalamus along rostral-caudal and superior-inferior axes. For example, Freesurfer's deep-learning-based algorithm has enabled greater accessibility of subunit-level segmentation (Fig. 3b).[30] However, most MRI-based techniques used in ALS neuroimaging studies employ 3 T machines, which lack the signal-to-noise ratio and spatial resolution necessary to accurately delineate individual hypothalamic nuclei. This contrasts with histological atlases and 7 T imaging (Fig. 3c–f), which allow for high-resolution visualisation of hypothalamic ultrastructure.[32]

The hypothalamus is a small structure (~5 *mm cross-section per hemisphere*), and its proximity to bone, major blood vessels, cerebrospinal fluid spaces, and air-filled sinuses contributes to imaging artefacts and signal distortion. These anatomical challenges are illustrated in Fig. 4. In addition, the spatial limitations of conventional 3 T T1-weighted sequences reduce anatomical precision, making subunit segmentation vulnerable to registration errors. Delineation is often based on anatomical landmarks rather than clear differences in tissue contrast or signal intensity.[33] Common parcellation schemes divide the hypothalamus into anterior, tuberal, and posterior subunits, reflecting early developmental patterning and rostrocaudal functional specialisation.[33]

The widely used segmentation framework proposed by Makris and colleagues was developed to overcome the limitations of 1.5 T imaging, but it conflates functional nuclei within and across subunits. For example, the PVN spans both the anterior-superior and superior-tuberal subunits, underscoring the difficulty of mapping functional nuclei onto anatomical parcellations.[33] These subunits are inferred based on overall morphology and depend heavily on registration accuracy and model regularisation. However, asymmetric nuclear atrophy—whether due to ageing or neurodegenerative disease—is often not well captured using these methods, owing to partial volume effects, limited signal-to-noise and contrast-to-noise ratios.[34] Furthermore, rater bias and training set composition can affect segmentation accuracy. In the case of Freesurfer's hypothalamic segmentation model, training was based on a small cohort of thirteen individuals, including both healthy controls and people with behavioural-variant frontotemporal dementia.[35] As a result, conclusions drawn about specific hypothalamic nuclei from subunit-based MRI studies should be interpreted with caution. Full anatomical validation of these segmentations requires histological correlation.[35]

Recent advances in neuroimaging—particularly 7 T MRI—have improved the capacity to resolve hypothalamic nuclei by offering higher resolution, superior contrast, and enhanced delineation of grey and white matter boundaries (Fig. 3e and f).[36,37] These benefits, however, are often accompanied by technical trade-offs, such as reduced field of view and lower signal-to-noise ratio in deep brain structures. As such, study protocols must be carefully optimised based on specific research objectives and anatomical targets of interest. Few ALS studies have incorporated 7 T MRI into their methodology.[38,39] As 7 T scanners become more widespread, they will enable more targeted and comprehensive interrogation of hypothalamic nuclei tailored to specific aspects of hypothalamic function.

structure and connectivity, these methods are limited in their ability to resolve functional impairments at the level of specific nuclei or neuronal populations. In this context, studies of hypothalamic-pituitary axis function—through neuroendocrine profiling or dynamic hormone testing—offer a valuable complementary approach. Assessing downstream hormonal output from axes such as the GH-IGF-1, HPA, HPT, and HPG systems may provide indirect but functionally meaningful evidence of hypothalamic disruption in ALS. These investigations could help localise dysfunction to specific hypothalamic circuits, refine our understanding of disease-related hypothalamic involvement, and identify clinically relevant biomarkers of metabolic impairment.[8] In Panel 3, we contextualise this in a summary of the clinical implications of hypothalamic dysfunction in ALS. Fig. 5 summarises key findings to date regarding hypothalamic–pituitary axis dysregulation in ALS.

Growth hormone/insulin growth factor-1 (GH-IGF1) axis

The GH–IGF-1 axis plays an essential role in somatic growth and metabolic regulation, including effects on brain and muscle development. Between half and two-thirds of plwALS may exhibit GH deficiency,[51,59] with reports of both reduced basal GH levels[60] and diminished GH response following stimulation with GHRH plus arginine.[51,59] GH secretion is pulsatile and regulated by hypothalamic neurons within the PeVN and ARC,[61] implicating possible hypothalamic involvement in altered GH dynamics. However, GH peak levels do not correlate with disease duration or motor function,[51,59] and so it is not clear how this might be clinically relevant.

IGF-1 levels, which are modulated by GH, show variable alterations in ALS, with studies reporting reductions[61] and increases.[62] This heterogeneity may reflect interindividual differences related to age, sex, and adiposity, and is compounded by small cohort sizes. In preclinical models, GH and IGF-1 levels initially rise after symptom onset before declining later in disease.[52] While GH axis abnormalities suggest hypothalamic dysregulation, peripheral factors such as ghrelin also modulate GH release. Reduced ghrelin levels reported in ALS[52] may contribute to impaired GH secretion.[63] Taken together, these findings support possible dysfunction in hypothalamic components of

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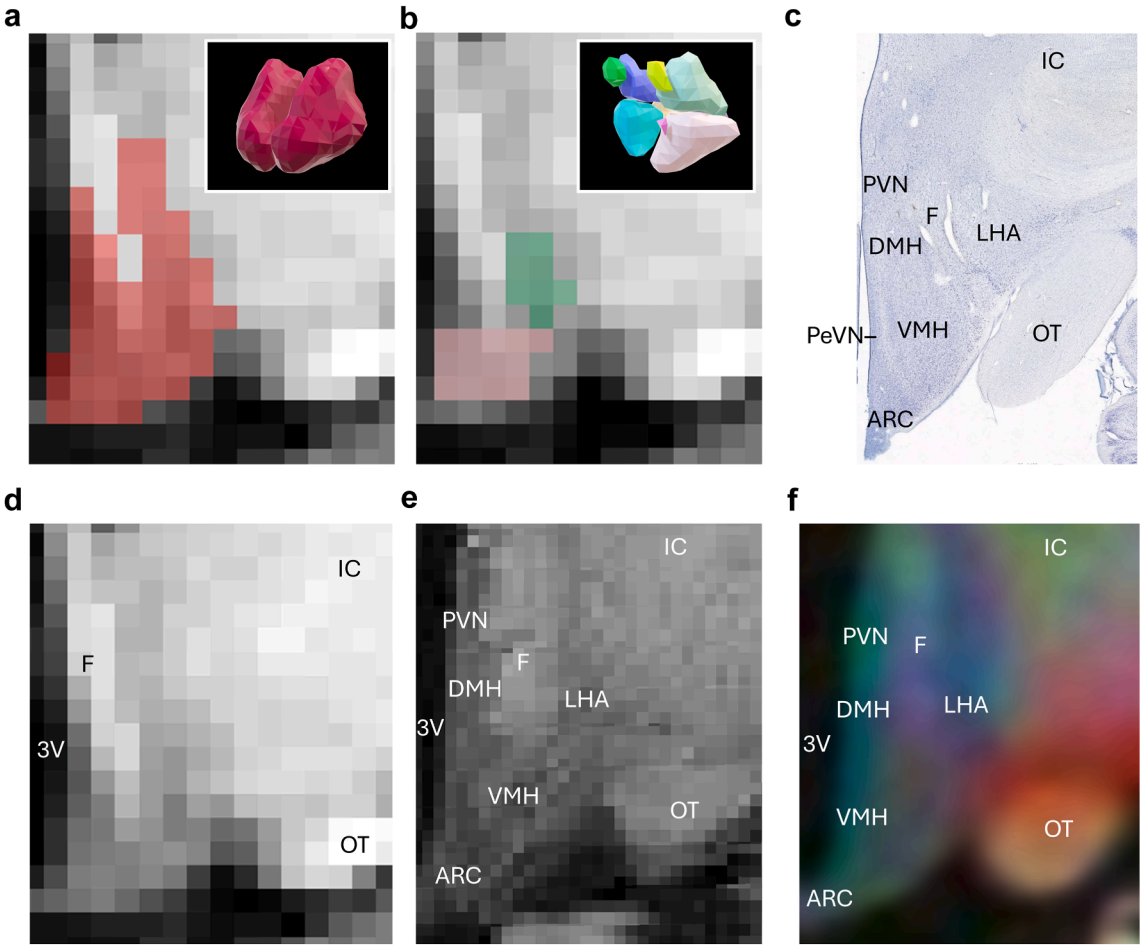

Fig. 3: **(a)** Whole-hypothalamus segmentation at 3 T using OSHy-X, with corresponding 3D model. **(b)** Subunit segmentation at 3 T using Freesurfer, with corresponding 3D model. Subunits visible include the left anterior-superior (lime green), right anterior-superior (green), left anterior-inferior (pink), left superior-tuberal (mint green), right superior-tuberal (blue), left inferior-tuberal (light pink), and right inferior-tuberal (light blue) regions. **(c)** Histological annotation of the hypothalamus on a Nissl-stained human section from the Allen Human Brain Atlas (atlas.brain-map.org). **(d)** T1-weighted MRI of the hypothalamus at 3 T. **(e)** High-resolution 7 T MRI with anatomical landmarks. Image courtesy of Professor Markus Barth, The University of Queensland, Australia. **(f)** Multi-contrast image from the HumanBrainAtlas that combines 7 T T1-weighted and diffusion-weighted images. *Abbreviations:* 3V = third ventricle; F = fornix; OT = optic tract; IC = internal capsule; ARC = arcuate nucleus; VMH = ventromedial hypothalamus; DMH = dorsomedial hypothalamus; PeVN = periventricular nucleus.

the GH-IGF-1 axis, though further work is needed to determine whether this reflects primary hypothalamic pathology or downstream effects.

Hypothalamic–pituitary–adrenal (HPA) axis

The HPA axis regulates stress responses and systemic homeostasis, and growing evidence suggests its dysregulation in ALS.[64] Corticotropin-releasing hormone (CRH), produced in the PVN, drives adrenal cortisol release. Postmortem studies show adrenal enlargement,[65] and elevated cortisol levels are reported in ALS, particularly in faster-progressing phenotypes.[53,54] Sleep disturbances, commonly observed in ALS, may be linked to both lateral hypothalamic dysfunction[26] and broader HPA axis involvement.[66] Circadian disruption of cortisol secretion and blunted cortisol awakening response have also been reported, with associations to depressive symptoms and disease severity.[54] Notably, oxytocin-producing neurons in the PVN appear reduced in ALS, whereas vasopressin neurons remain intact.[26] As oxytocin inhibits cortisol release, this may represent a relevant mechanism underlying HPA dysregulation. In contrast, data on upstream CRH and adrenocorticotropin hormones in ALS are sparse. While strong links between glucocorticoid dysregulation and neurodegeneration exist in other conditions,[66] further research is needed to establish the role of the HPA axis in ALS pathophysiology.

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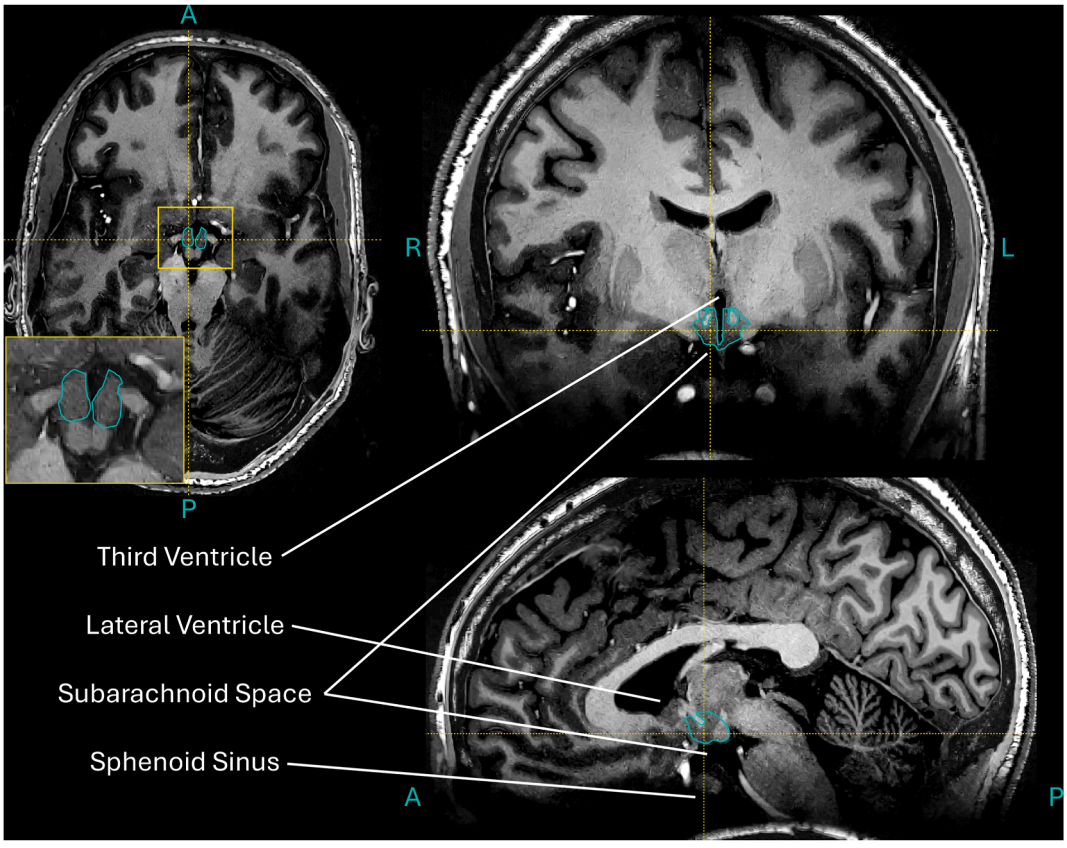

Fig. 4: The hypothalamus (delineated in teal) is surrounded by adjacent structures that can introduce imaging artefacts in magnetic resonance studies. Image shown is a 7 T scan from a healthy male in their 30s. Image courtesy of Professor Markus Barth, The University of Queensland, Australia. *Abbreviations:* A = anterior; P = posterior; L = left; R = right.

Hypothalamic–pituitary–thyroid (HPT) axis

Thyroid hormone (TH) regulates thermogenesis and basal metabolic rate, making it relevant to energy imbalance in ALS. Early studies offered limited evidence of overt thyroid dysfunction in ALS,[67] though lower thyrotropin-releasing hormone (TRH) levels in the spinal cord[57] and reduced TRH receptor expression have been reported.[68] TRH is produced in the PVN and is the primary hypothalamic regulator of the HPT axis.[69] More recent studies found no association between TH levels and survival,[58] and so as with studies in GH, clinical relevance remains unclear. In the 1980s, TRH was trialled as a potential treatment for ALS, based on its role in other neurological diseases such as spinocerebellar ataxia.[70] However, clinical outcomes were inconsistent and ultimately inconclusive.[69] At present, there is little evidence that HPT axis dysfunction plays a central role in ALS. Nonetheless, subtle alterations in HPT regulation could affect mitochondrial energetics or thermoregulation in a subset of patients, warranting further study.

Hypothalamic–pituitary–gonadal (HPG) axis

Sex differences in ALS—particularly in incidence, progression, and prognosis—have prompted investigation of the HPG axis.[71] Males generally have worse outcomes, with more rapid weight and function loss.[56] Lower levels of dihydrotestosterone[55] and free testosterone[72] have been reported in male patients, though HPG axis integrity appears preserved based on hormonal stimulation studies.[73] These findings must also be considered in the context of androgen receptor mutations in spinal and bulbar muscular atrophy, another X-linked MND.[56] Testosterone may have neuroprotective properties in ALS, with evidence supporting motor unit preservation in both human and animal studies.[74,75] In females, ALS incidence increases postmenopause,[56] and exposure has been associated with improved survival.[76] Hormone replacement therapy has been linked to reduced ALS risk in some cohorts.[77] Data on progesterone are mixed, with both elevated[53] and reduced levels reported.[75]

Sex hormones strongly influence metabolism, and sex-specific metabolic profiles may partly explain differences in ALS outcomes. Testosterone promotes muscle mass and energy expenditure, while oestrogen supports mitochondrial function and lipid metabolism. Indirectly, these hormones may also impact appetite and energy intake—for example, 17α-oestradiol (non-

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Panel 3: **Clinical implications of hypothalamic dysfunction in amyotrophic lateral sclerosis.**

Emerging evidence suggests that hypothalamic dysfunction in amyotrophic lateral sclerosis (ALS) may contribute to a range of non-motor symptoms with direct clinical consequences, supporting the concept of ALS as a multisystem disease (Table 1). The hypothalamus plays a central role in regulating energy homeostasis, appetite, circadian rhythms, and endocrine function—domains that are increasingly recognised as clinically relevant in ALS. Disruption of these systems may exacerbate disease progression and affect quality of life, yet they remain under-recognised in routine clinical care.

Weight loss and metabolic dysregulation

Hypothalamic atrophy—particularly in the anterior and tuberal regions—has been associated with accelerated weight loss, lower body fat mass, and reduced survival in ALS. These changes may reflect impaired satiety signalling, altered energy expenditure, or dysregulated hormonal pathways. Preserved fat mass is linked to better outcomes, and identifying individuals at risk of hypothalamic-driven metabolic decline may guide early nutritional or pharmacological interventions.

Appetite disturbance and feeding behaviour

Loss of appetite in ALS is multifactorial but may reflect disrupted hypothalamic circuits rather than solely dysphagia or depression. Imaging and postmortem studies implicate neuropeptidergic neurons (e.g., POMC, MCH, orexin) in appetite regulation. Appetite loss is associated with poorer nutritional status and may represent a modifiable contributor to disease trajectory.

Neuroendocrine and hormonal dysregulation

Dysfunction across multiple hypothalamic–pituitary axes has been observed in ALS. These include:

- *Growth Hormone/Insulin Growth Factor-1 (GH-IGF1) axis:* Possible hypothalamic impairment; GH deficiency is common and may affect muscle and metabolic maintenance.
- *Hypothalamic-Pituitary-Adrenal (HPA) axis:* Altered cortisol rhythms and stress response; may contribute to sleep disruption and depressive symptoms.
- *Hypothalamic-Pituitary-Gonadal (HPG) axis:* Sex hormone differences may influence disease risk, progression, and metabolism, with emerging relevance for sex-specific care.
- *Hypothalamic-Pituitary-Thyroid (HPT) axis:* Role remains uncertain but may contribute to altered thermogenesis and energy balance in select patients.

Cognitive and behavioural symptoms

Reductions in hypothalamic volume have been linked to cognitive impairment and may contribute to behavioural symptoms across the ALS-FTD spectrum. Hypothalamic involvement in sleep-wake regulation and circadian timing may also underlie fatigue and sleep disturbances in ALS.

Clinical opportunities and unmet needs

Despite growing recognition, hypothalamic dysfunction remains under-assessed in ALS. There is a need for validated clinical tools and biomarkers to detect hypothalamic dysfunction early, such as neuroimaging, endocrine profiling, or appetite assessments. Interventions targeting hypothalamic pathways—like ghrelin agonists, MCH modulation, and metabolic therapies—may hold therapeutic potential but remain to be evaluated in prospective trials.

feminising oestrogen) stimulates appetite, acting via hypothalamic POMC-expressing neurons.[78] These differences may modulate ALS progression—males may face greater energetic deficits due to higher lean mass and resting energy requirements, whereas females may benefit from higher fat reserves. Substrate utilisation

Hypothalamic feature or axis	Observed dysfunction	Associated clinical manifestations	Evidence base	Clinical relevance
Structural atrophy (MRI)	Volume loss (especially anterior, superior, posterior subunits)	Weight loss, cognitive impairment, reduced survival	Gorges 2017[42]; Chang 2023[38]; Michielsen 2024[45]	Prognostic marker; risk stratification
Arcuate-paraventricular nuclei circuitry	Loss of POMC, NPY/AgRP signalling balance	Appetite loss, weight loss, altered satiety	Bolborea 2023[13]; Gabery 2021[14]	Potential therapeutic target (e.g., ghrelin)
Lateral hypothalamic area dysfunction	MCH/orexin neuron loss or dysfunction	Appetite changes, arousal/sleep disturbance	Guillot 2025[26]; Gabery 2021[14]	Targetable via MCH/orexin pathways
GH-IGF1 axis	GH deficiency; altered IGF-1 levels	Muscle atrophy, reduced regeneration, metabolic decline	Saccà 2012[51]; Steyn 2013[52]	Hormonal biomarker; experimental interventions
HPA axis	Elevated cortisol; blunted rhythm	Stress sensitivity, depression, sleep disruption	Gargiulo 2011[53]; Spataro 2015[54]	Possible driver of neuropsychiatric symptoms
HPG axis	Sex hormone dysregulation	Sex-specific progression rates and metabolic profiles	Sawal 2020[55]; Grassano 2024[56]	Relevance for personalised care
HPT axis	Limited evidence of Thyroid hormone or thyrotropin-releasing hormone dysfunction	Thermoregulation; energy expenditure	Mitsuma 1984[57]; Zheng 2014[58]	Unclear; may affect metabolic reserve

Table 1: **Summary of hypothalamic dysfunction in ALS and its clinical consequences.**

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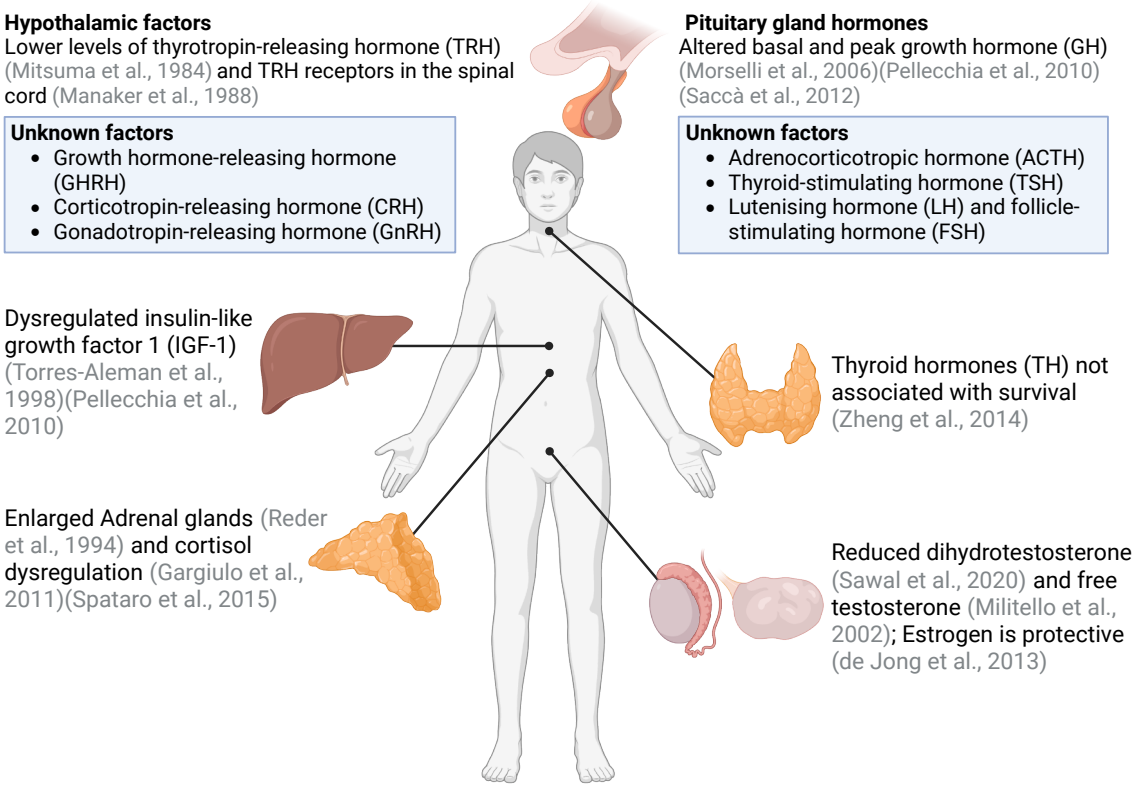

Fig. 5: Summary of evidence for dysregulation across the hypothalamic–pituitary axes in amyotrophic lateral sclerosis (ALS).

differences (e.g., lipid vs carbohydrate metabolism) may further shape disease trajectories. Despite their relevance, these sex-specific metabolic effects remain underexplored in ALS research. Finally, we note that upstream regulation of the HPG axis—such as pulsatile GnRH secretion—also remain unexplored.

Conclusion

Emerging evidence highlights the hypothalamus as a key site of involvement in ALS, with implications for the regulation of energy homeostasis, sleep, cognition, and behaviour—domains increasingly recognised as clinically relevant in this disease. Histopathological and imaging studies reveal specific changes in hypothalamic structure, including selective neuronal loss, subregional atrophy, and widespread proteinopathy. These findings suggest that hypothalamic dysfunction may contribute to the non-motor symptom burden and metabolic disturbances that influence disease trajectory and prognosis in plwALS.

Outstanding questions

Despite a growing body of evidence, the functional consequences of hypothalamic pathology in ALS remain only partially understood. The hypothalamus regulates diverse autonomic and endocrine processes, and its disruption could contribute to the heterogeneity of clinical presentations seen in ALS. Clarifying the downstream physiological effects of hypothalamic dysfunction—particularly on appetite regulation, body composition, and stress responses—may reveal new opportunities for intervention. Moreover, how might hypothalamic dysfunction contribute to sleep disturbance in ALS, and what does this reveal about its broader multisystem effects? Future research should prioritise multimodal approaches that integrate histological, neuroimaging, and endocrine data to map hypothalamic involvement more precisely. Systems biology and omics-based techniques may help identify molecular targets for intervention, while advanced neuroimaging platforms, including ultra-high-field MRI (i.e., 7 T), will enable *in vivo* characterisation of hypothalamic nuclei and their functional connectivity. These advances could support the development of biomarkers and targeted therapies aimed at modulating hypothalamic pathways, particularly in patients at high risk of rapid disease progression.

Ultimately, addressing hypothalamic dysfunction in ALS may offer new strategies to improve clinical

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Search strategy and selection criteria

References for this Review were identified by searches of PubMed covering the full range of available publication dates up to March 2025, as well as by reviewing references from relevant articles. The search terms included "Amyotrophic Lateral Sclerosis", "ALS metabolism", "hypothalamus ALS", "energy homeostasis", "neuroendocrine dysfunction", "ALS weight loss", "hypermetabolism ALS", "hypothalamic atrophy", and "hypothalamic-pituitary axis dysfunction". There were no language restrictions. The final reference list was selected based on relevance to the topics covered in this Review, prioritising studies providing mechanistic insights, clinical correlations, and translational potential.

outcomes and quality of life. A more nuanced understanding of hypothalamic involvement may also help explain clinical variability, especially when considering disease progression. This will help support the development of personalised treatment approaches tailored to individual patient needs and the burden associated with non-motor aspects of the disease.

Contributors

JC–Conceptualisation, visualisation, writing original draft, writing review & editing; AHT–Conceptualisation, writing original draft, writing review & editing; TBS–Writing original draft, writing review & editing; LD–Writing review & editing; STN–Conceptualisation, supervision, writing review & editing; FJS–Conceptualisation, project administration, supervision, writing original draft, writing review & editing. All authors read and approved the final version of the manuscript.

Declaration of interests

The authors report no competing interests.

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